

Sentinelleai — Audit Report

MagnetaLpAtomicHelper.sol · 5132e4b3-9fdd-4a22-a98e-242414ceed4b · 2026-06-12 15 h
09 min 33 s

PASS 88 / 100 2 consolidated findings

MagnetaLpAtomicHelper.sol presents a well-structured, minimal-surface atomic LP helper with no critical or high-severity vulnerabilities identified across five independent auditors. The contract correctly applies ReentrancyGuard (SC01), SafeERC20, ephemeral approvals with post-call revocation, slippage floors on addLiquidity (SC04), and a deadline buffer guard. No persistent state, no ETH handling, no delegatecall, and no privileged roles reduce the attack surface substantially. The only findings are low-severity code quality issues: inconsistent use of custom errors versus require-string patterns in the For-variant entry points, and a slightly misleading NatSpec comment on the deadline error. Neither finding triggers an OWASP SC Top 10 category at meaningful severity. The contract is suitable for production deployment with the minor recommendations applied.

Severity breakdown

1 LOW 1 INFO

Consolidated findings

LOW CVSS 2.1 SC09 - Code Quality / Gas Inefficiency

Inconsistent Error Reporting: Custom Errors vs. require Strings

The contract defines and uses custom errors (DeadlineTooClose, LpAmountZero, NewLpZero, ZeroAddress) throughout the private logic, but the public entry points `compoundPositionFor` and `migratePositionFor` fall back to `require()` with string messages for the zero-recipient check. This is inconsistent with the apparent design intent, wastes gas on revert-string encoding, and may confuse off-chain tooling that parses revert reasons.

```
compoundPositionFor (line ~134): require(recipient != address(0), ...);  
migratePositionFor (line ~251): require(recipient != address(0), ...)
```

→ Add a `ZeroRecipient` custom error and replace both `require` statements with: `if (recipient == address(0)) revert ZeroRecipient();`

Confirmed by: developer

INFO CVSS 0.5 SC09 - Documentation Accuracy

Misleading NatSpec on DeadlineTooClose Error Semantics

The NatSpec comment for `DeadlineTooClose` states it 'does NOT mean the deadline already passed', but the guard condition (`deadline <= block.timestamp + MIN_DEADLINE_BUFFER`) also catches already-expired deadlines (`deadline < block.timestamp`). The comment is technically inaccurate and could mislead integrators parsing revert reasons.

Error declaration and NatSpec comment, lines ~66-70

→ Update the comment to: 'Reverted when the supplied deadline is at or within `MIN_DEADLINE_BUFFER` seconds of the current block, including already-expired deadlines.'

Confirmed by: developer

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This report combines deterministic regex+AST sentinels, optional Slither/Mythril static analysis, and a 6-model adversarial LLM panel consolidated by a Judge agent.